Sub-1B VLM for Document Understanding: A Systematic Evaluation and a Targeted Improvement

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Abstract

Document understanding is a natural target for on-device vision-language models (VLMs), yet models with fewer than a billion parameters remain largely uncharacterized on it. We present a systematic and reproducible evaluation of six open-source VLMs, ranging from 0.45 to 0.9B parameters, across five document benchmarks arranged as a diagnostic ladder that runs from recognition through extraction, numeric reasoning, and layout reasoning to tabular reasoning. Under matched decoding, precision, and sampling, the strongest model, Qwen3.5-0.8B, reaches a mean score of 70.3 out of 100 and leads on three of the five benchmarks, yet no model clears 50.1 on visual table question answering. Two results stand out. Within this size band, architecture matters more than parameter count: the smallest model, at 0.45B, outranks both 0.9B models. And the design choice that drives performance is the image-resolution strategy rather than scale. We then ask where the best model falls short. Breaking the table benchmark into its sub-domains shows that the weakness lies in Wikipedia-style visual lookup, at 27.8 and 33.1, rather than in financial or numeric tables, at 84.0, which corrects a common assumption. Guided by this evidence, we design a parameter-efficient adaptation aimed at the measured weakness, with strict separation between training and evaluation data and a success criterion fixed in advance. We release an extensible evaluation pipeline, a public training dataset, and all scripts. Our first adaptation does not meet that criterion: it induces catastrophic forgetting, which we trace to a mismatch between the answer style of the training data and that of the benchmarks, and we report the outcome in full (Section 8.1). The criterion rejecting the degraded checkpoint is the protocol behaving as intended, and the diagnosis points to a concrete corrected recipe.

1 Introduction

Most multimodal evaluation effort targets models of 3B parameters and above. This leaves open a practical question about the smaller models that actually fit on-device and cost-sensitive deployments: can a VLM below 1B parameters serve as a reliable foundation for document understanding and later domain adaptation? Document images are dense and high-resolution, so reading them stresses the vision encoder and the resolution-handling strategy far more than natural-image captioning does. The central question of this work is therefore:

> *Can sub-billion-parameter VLMs be a reliable foundation for document-understanding domain adaptation, and where do they break?*

We answer it in two parts. Part 1 is a controlled comparison of six models on a five-benchmark suite built as a diagnostic ladder rather than a leaderboard. Part 2 turns the resulting failure profile of the best model into a concrete, literature-grounded improvement strategy whose success criterion is fixed before any training. Our contributions are the following.

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- **An evaluation pipeline and a controlled comparison protocol.** A single config-driven entry point scores any registered model. We add custom integrations for three models the harness does not support out of the box (FastVLM-0.5B, Qwen3.5-0.8B, and LFM2.5-VL-450M), and we hold the comparison fair and fast by aligning every model to 16-bit precision, so none benefits from a wider numeric format, and by capping generation length in a way that leaves accuracy unchanged while cutting decoding cost (Section 5).
- **A reproducible six-model evaluation** across the five document benchmarks under this protocol, with a per-skill analysis and a joint view of accuracy and efficiency (Sections 4 to 6).
- **Two robust findings:** architecture beats parameter count between 0.45 and 0.9B, and accuracy tracks the resolution and tokenization strategy more closely than it tracks scale (Section 6).
- **An evidence-based gap analysis** that decomposes the table benchmark into its sub-domains and shows the deficit is visual row-and-column lookup rather than numeric reasoning, contrary to the default assumption in the table-QA literature (Section 7).
- **A targeted, contamination-controlled adaptation** of the best model at the measured weakness, evaluated against a pre-registered no-regression criterion, together with a publicly released training dataset (Section 8).

2 Related Work

Small and edge VLMs. Efficiency-focused models handle the vision pathway in noticeably different ways to fit tight compute budgets. SmolVLM couples a SigLIP encoder [20] with a small language backbone and aggressive visual-token compression [13]. LLaVA-OneVision uses AnyRes multi-crop tiling [10]. The Qwen-VL line adopts native dynamic-resolution processing [19] in the spirit of NaViT [3], and InternVL pairs a stronger vision encoder with dynamic high-resolution tiling [2]. Each design answers the two questions that govern document performance differently, namely how to encode a high-resolution page and how much language capacity backs the reading, which is exactly why we assemble a diversified shortlist rather than testing one recipe repeatedly.

Document-understanding benchmarks. We build on established public benchmarks: DocVQA for dense scanned documents [15], InfographicVQA for high-resolution layout reasoning [16], ChartQA for chart-based numeric reasoning [14], OCRBench for raw text recognition [12], and TableVQA-Bench for visual table question answering [9]. We evaluate through VLMEvalKit [5], a generation-based harness with per-benchmark scorers.

Parameter-efficient adaptation and table understanding. LoRA [7] and its quantized variant QLoRA [4] allow low-cost specialization while limiting catastrophic forgetting. For tables in particular, Table-LLaVA and the MMTab corpus show that table-structure-aware supervision is what unlocks visual tabular reasoning [21]. Part 2 builds on this line of work, but aims at a sub-skill we have actually measured rather than at tables in the aggregate.

3 Models

Selection criteria. We require each candidate to meet four conditions: a verified total parameter count below 1B including the vision encoder; a generalist question-answering model, which excludes OCR and parsing specialists that emit transcriptions rather than answers; support for document-scale input resolution; and a real, loadable checkpoint. Among the popular generalist VLMs that satisfy these conditions, we pick one strong representative per architecture and lab, so the comparison isolates design differences instead of re-testing one recipe. The resulting six models (Table 1) span six labs, five encoder paradigms, and four language backbones. Most important for our purposes, they also span distinct resolution strategies, the axis that matters most for dense pages.

Table 1: The six evaluated sub-1B VLMs. Parameter counts include the vision encoder. Resolution strategy is the primary design axis under study.

Model	Lab	Params	Encoder + backbone	Resolution strategy
Qwen3.5-0.8B [18]	Alibaba	0.8B	Qwen-native + gated-delta/MoE	Native dynamic (NaViT-style)
InternVL3-1B [17]	OpenGVLab	0.9B	InternViT-300M + Qwen2.5-0.5B	Dynamic high-res tiling (448^2)
LFM2.5-VL-450M [11]	Liquid AI	0.45B	SigLIP2-86M + LFM2-350M	Native $\leq 512^2$ + tunable tiling
LLaVA-OV-0.5B [10]	LMMs-Lab	0.9B	SigLIP + Qwen2-0.5B	AnyRes multi-crop
SmolVLM2-500M [8]	Hugging Face	0.5B	SigLIP + SmolLM2-360M	Split + pixel-shuffle compression
FastVLM-0.5B [1]	Apple	0.8B	FastViTHD + Qwen2-0.5B	High-res, few-token compression

Table 2: The five document benchmarks, ordered as a capability ladder. Metrics are native to each benchmark. ANLS is average normalized Levenshtein similarity; relaxed accuracy admits $\pm 5\%$ numeric error.

Benchmark	Image type	Core sub-skill	Metric
OCRBench	Mixed scenes/scans/handwriting (5 tasks)	Raw text recognition	Accuracy
DocVQA (val)	Scanned industry documents	Read + locate (single-hop)	ANLS
ChartQA (test)	Real charts (bar/line/pie)	Data extraction + arithmetic	Relaxed acc.
InfoVQA (val)	High-res infographics	Text + graphics + layout	ANLS
TableVQA-Bench	Rendered table images (4 sub-domains)	Tabular lookup + aggregation	Accuracy

4 Benchmark Suite and Metrics

A diagnostic ladder. We do not treat the five benchmarks as interchangeable accuracy numbers. Instead we order them so that each asks for more than the last (Table 2): recognition with OCRBench, then single-hop extraction with DocVQA, numeric reasoning with ChartQA, multi-hop layout reasoning with InfoVQA, and finally structured tabular reasoning with TableVQA. OCRBench anchors the ladder. When a model fails a task higher up, its recognition score tells us whether the problem is reading or reasoning, and that distinction is what lets the Part-2 gap analysis rest on evidence rather than speculation. All five benchmarks are scored locally, which keeps the pipeline reproducible.

Beyond-accuracy evaluation. The suite is deliberately diverse in its metrics. It uses three different notions of correctness (accuracy, ANLS, and relaxed accuracy), so tolerance to exact string matching is built into the comparison from the start. We also report per-sample latency, a deployment-relevant axis (Section 6), which lets us describe each model by accuracy and efficiency together rather than by a single ranking. The task brief names two further axes beyond accuracy, calibration and robustness to domain terminology. We have not yet measured them. Instead we give a precise protocol for both in Section 9 and treat them as extensions to run once the baseline results warrant it, not as speculative additions.

5 Experimental Setup and Reproducibility

Inference protocol. All models use greedy decoding (`do_sample` = False). We cap generation at `max_new_tokens`=128 for two reasons. It keeps the comparison fair, since no model is penalized or rewarded for a runaway transcript, and it keeps the run fast, because these small models otherwise spend their decoding budget on verbose restatements. An ablation (Section 5.1) confirms the cap leaves accuracy unchanged on these short-answer benchmarks. Each benchmark is evaluated on a fixed, seed-42 stratified subset of N=1000 questions (OCRBench has exactly 1000 and is run in full), drawn deterministically so that the numbers reproduce and the per-sub-domain composition of TableVQA is preserved. To keep numeric precision from confounding the comparison, we align every model to 16-bit, the format used in realistic edge deployment, so that none gains an advantage from a wider representation. We use fp16 for SmolVLM2, LLaVA-OV, and FastVLM, and bf16 for InternVL3, Qwen3.5, and LFM2.5-VL, and we document the per-model choice because forcing a bf16-native model into the narrower range of fp16 can destabilize it.

Table 3: Inference-efficiency ablations (SmolVLM2-500M, $N{=}100$). Each lever is evaluated for both speed and accuracy. The fp16 score change is within precision/sampling noise at this N; the generation cap is exactly neutral.

Lever	Benchmark	Setting	Score	Gen. time (s)
Precision	OCRBench	fp32	57.0	209
		fp16	62.0	**116**
Gen. cap	DocVQA	2048 tok	76.6	≈138
		128 tok	76.6	
	InfoVQA	2048 tok	32.9	
		128 tok	32.9	

Table 4: Accuracy (0 to 100; higher is better) on the five document benchmarks, $N{=}1000$ seed-42 subsets. Sorted by mean. **Bold** = best per column.

Model	Params	ChartQA	DocVQA	InfoVQA	OCRBench	TableVQA	**Mean**
Qwen3.5-0.8B	0.8B	70.8	**89.3**	**62.3**	79.2	**50.1**	**70.3**
InternVL3-1B	0.9B	68.8	80.8	54.2	**79.4**	33.5	63.3
LFM2.5-VL-450M	0.45B	**73.1**	77.2	41.3	67.8	40.2	59.9
LLaVA-OV-0.5B	0.9B	60.0	70.4	39.0	60.2	34.4	52.8
SmolVLM2-500M	0.5B	60.4	67.8	27.4	61.0	30.3	49.4
FastVLM-0.5B	0.8B	44.3	63.6	32.5	26.2	17.5	36.8

Software stack and pipeline. We use VLMEvalKit [5] as a pinned dependency that we never fork. It handles dataset download, prompt construction, generation, and per-benchmark scoring, and adding a model to it requires implementing a single `generate_inner` method. Around it we build a thin, config-driven runner in which a registry maps a model name to a loader, so that one command, `run_eval.py --models <name>`, scores any registered model on any subset with per-pair caching and resumption. One practical contribution is extending the harness to three models it does not cover out of the box: we write device-agnostic wrappers for FastVLM-0.5B and Qwen3.5-0.8B, both of which need trust-remote-code-style loading and bespoke image-token handling, and add a registry entry for LFM2.5-VL-450M, so that all six models run under one identical protocol. The engine commit, seeds, and decoding parameters are pinned, and results persist to a single output tree that the aggregation and figure scripts consume.

Hardware. Evaluation ran on free or low-cost Colab GPUs (NVIDIA T4 and L4), with the Part-2 study on an A100. Because decoding is deterministic, the accuracy numbers do not depend on the hardware; latency does, so we treat it as indicative (Section 10).

5.1 Efficiency ablations

Two protocol choices trade compute for throughput, and we justify each with a controlled ablation that reports its effect on accuracy, not only its speedup (Table 3; SmolVLM2-500M, $N{=}100$). **Precision.** Downcasting the one fp32 model to fp16 gives a $1.8\times$ speedup with no instability. The $+5.0$-point score change at $N{=}100$ sits within run-to-run precision variation at this sample size and shrinks at $N{=}1000$, so we read the change as accuracy-neutral and simply faster. **Generation cap.** Lowering `max_new_tokens` from 2048 to 128 leaves the DocVQA and InfoVQA scores bit-identical and the wall-time unchanged, because these short-answer models emit end-of-sequence well before the cap. The cap therefore truncates no correct answer while removing the tail-latency risk of runaway generation. Both ablations are reproduced by `efficiency_ablation.py`.

6 Results and Analysis

Table 4 reports accuracy across the suite. Figure 1 shows the per-benchmark profile, and Figure 2 places the models on the accuracy and efficiency frontier.

Table 5 reports the per-sample latency underlying the efficiency axis of Figure 2.

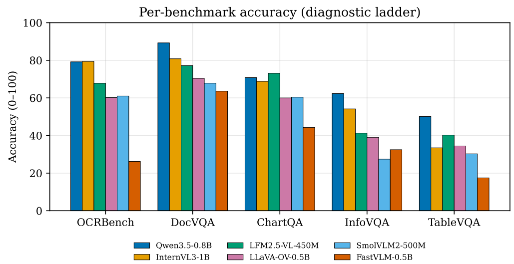

Figure 1: Per-benchmark accuracy across the diagnostic ladder. The spread widens from recognition to layout/table reasoning.

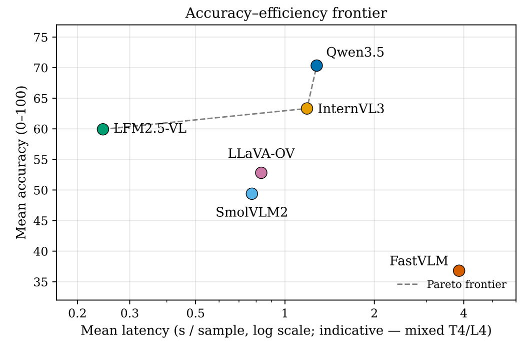

Figure 2: Accuracy against efficiency. LFM2.5-VL and Qwen3.5 own the Pareto front, and FastVLM is dominated. Latency is indicative (mixed T4/L4).

Table 5: Mean per-sample latency (seconds; lower is better). Indicative only, since the T4 and L4 hardware differs across models, so only the ordering is robust (Section 10).

Model	ChartQA	DocVQA	InfoVQA	OCRBench	TableVQA	Mean
LFM2.5-VL-450M	0.14	0.27	0.23	0.32	0.27	**0.24**
SmolVLM2-500M	0.72	0.96	0.70	0.83	0.66	0.77
LLaVA-OV-0.5B	0.46	1.35	0.83	0.95	0.57	0.83
InternVL3-1B	0.69	1.47	1.84	1.46	0.47	1.19
Qwen3.5-0.8B	0.57	1.44	1.49	2.22	0.67	1.28
FastVLM-0.5B	3.84	3.92	3.92	3.68	3.93	3.86

Per-benchmark analysis. On **OCRBench**, Qwen3.5 (79.2) and InternVL3 (79.4) lead, the other capable models cluster between 60 and 68, and FastVLM collapses to 26.2 despite its high-resolution encoder. Its aggressive few-token compression appears to trade away exact-reading fidelity. **DocVQA** shows the tightest spread among the capable models, because ANLS tolerates minor reading slips; FastVLM even recovers to 63.6 here, which tells us its OCRBench failure is about exact-match reading rather than comprehension. **ChartQA** is the least encoder-bound task, and the smallest model wins it (LFM2.5-VL, 73.1). **InfoVQA** is the most discriminating task, spanning 27.4 to 62.3, where native dynamic resolution (Qwen3.5, InternVL3) clearly beats aggressive token compression (SmolVLM2). On **TableVQA** every model scores low, and Qwen3.5's 50.1 is the ceiling.

Key findings. **A clear best foundation.** Qwen3.5-0.8B reaches the highest mean (70.3) and wins three of five tasks. Its native dynamic-resolution encoding, paired with the largest backbone in the set, generalizes across reading, layout, and tables. **Architecture beats parameter count.** The smallest model, LFM2.5-VL-450M at 0.45B, outranks both 0.9B models and wins ChartQA, so within this band the resolution strategy and backbone quality matter more than scale. **Resolution strategy orders the field.** Native dynamic resolution and tunable tiling come first, AnyRes multi-crop next, and aggressive token compression last; FastVLM's few-token compression is an outlier liability for exact reading. **Efficiency follows architecture, not size.** The image-token budget dominates per-sample cost, so the 0.45B LFM2.5-VL runs about five times faster than the 0.8B FastVLM, and the frontier of accuracy against efficiency belongs to LFM2.5-VL, the cheapest, and Qwen3.5, the most accurate. **Tables are the frontier** for every model, which sets up Part 2.

7 Knowledge-Gap Analysis

The best model, Qwen3.5-0.8B, reads well (OCRBench 79.2, DocVQA 89.3) and handles layout well (InfoVQA 62.3), but it reaches only 50.1 on visual table question answering. That is its weakest task, 12 points below its next-lowest score and 39 points below its DocVQA. Tables are therefore the decisive bottleneck. To understand why, we break TableVQA-Bench into its four native sub-domains (Table 6, Figure 3).

Table 6: TableVQA-Bench sub-domain accuracy for Qwen3.5-0.8B. The deficit is concentrated in Wikipedia-style visual lookup (VWTQ/VWTQ-Syn), the largest slice of the benchmark; financial tables are already strong.

Sub-domain	What it tests	Accuracy
VWTQ	Lookup over real Wikipedia-style tables (largest slice)	**27.8**
VWTQ-Syn	Lookup over synthetic Wikipedia-style tables	**33.1**
VTabFact	True/false fact verification	55.5
FinTabNetQA	Financial tables (numbers/units)	84.0

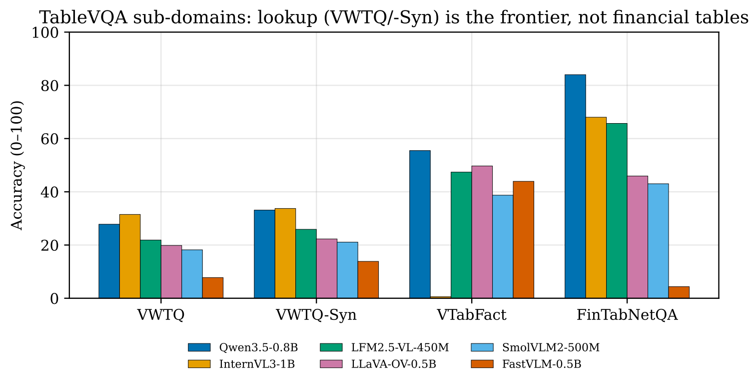

Figure 3: TableVQA sub-domain accuracy for all six models. Lookup (VWTQ/VWTQ-Syn) is the universal floor; financial tables (FinTabNetQA) are the universal ceiling. The aggregate "table" deficit is a *visual lookup* deficit, not a numeric-reasoning one.

The decomposition is decisive. Qwen3.5 is already strong on financial, number-and-unit tables (84.0) and reasonably strong on fact verification (55.5), while its failures concentrate on plain lookup over dense, general-purpose tables (27.8 and 33.1). Since VWTQ makes up roughly half of the benchmark, this sub-skill also carries the most weight in the aggregate score. The result runs against the usual assumption in the table-QA literature, which treats financial and numeric tables as the hard case, and it sharpens the diagnosis: the model can read the text in a table but struggles to bind a question to the correct row and column when the layout is busy. We therefore target visual table lookup rather than numeric reasoning.

8 Improvement Strategy and Proof of Concept

Method. Guided by the diagnosis, we apply a light parameter-efficient adaptation to Qwen3.5-0.8B that targets visual table lookup while protecting the four capabilities the model is already good at. We use bf16 LoRA [7], which is known to learn less and forget less and is therefore the safe choice for a strong baseline we must not regress, and we avoid 4-bit quantization [4] because the model's gated-delta state updates are sensitive to it and the 0.8B model fits in bf16 anyway. We freeze the vision encoder and place the adapters on the language pathway, including the gated-delta projections that make up most of the model's sequence-mixer layers, so that adaptation does not disturb the visual features the other benchmarks rely on.

Data, with strict isolation. The training data is weighted toward Wikipedia-style visual table lookup, the sub-skill we measured as weakest, and mixed with about 25% replay of the model's already-strong document data to guard against catastrophic forgetting, a known risk when a strong model is specialized on a narrow task. Two safeguards keep training and evaluation cleanly separated. Replay is drawn only from rows outside the fixed seed-42 evaluation subsets, and every candidate training question is checked against the full TableVQA-Bench question set and dropped on a match. TableVQA-Bench is never itself a training source. The table-QA prompts reuse the harness's own template, so the model trains under the same prompt it is scored under. We release the resulting

Table 7: The targeted adaptation *fails* the pre-registered gate, on the identical seed-42 subsets: the target task itself regresses and the strong DocVQA/InfoVQA capabilities collapse.

Metric	Baseline	After adapt.	Δ
TableVQA (mean)	50.1	19.6	-30.5
VWTQ	27.8	0.6	-27.2
VWTQ-Syn	33.1	0.0	-33.1
VTabFact	55.5	62.4	$+6.9$
FinTabNetQA	84.0	15.4	-68.6
OCRBench	79.2	77.4	-1.8
DocVQA	89.3	50.8	-38.4
ChartQA	70.8	75.3	$+4.5$
InfoVQA	62.3	27.8	-34.5

mixture as a public dataset with a manifest recording these safeguards.[1] This design follows the evidence that table-structure-aware supervision is the effective lever for visual tabular reasoning [21].

Pre-registered evaluation. We fix the success criterion before training. The adaptation is accepted only if the TableVQA mean rises by at least 3 points, with the gain concentrated in the targeted VWTQ sub-domains, and no protected benchmark (OCRBench, DocVQA, ChartQA, InfoVQA) drops by more than 1.5 points, all measured on the identical seed-42 subsets. Because TableVQA-Bench is small, about 1500 items, we always report the sub-domain breakdown rather than the mean alone.

Expected outcomes. We expect a modest gain focused on VWTQ and VWTQ-Syn, with the protected benchmarks staying within tolerance. We do not expect the gap to close fully, because at 0.8B the vision encoder limits how much of a dense table is legible. A measured lift on the highest-leverage sub-skill without regression would show that the localized gap is at least partly addressable by targeted PEFT, which is the practical claim of Part 2.

8.1 Improvement results: a rejected adaptation

The targeted adaptation does not pass the pre-registered criterion (Table 7). Instead of the modest gain we anticipated on visual lookup, TableVQA drops from 50.1 to 19.6, and two benchmarks the model had handled well collapse (DocVQA from 89.3 to 50.8, InfoVQA from 62.3 to 27.8). The criterion therefore rejects this checkpoint, which is exactly what a no-regression protocol is meant to do: it keeps us from shipping a degraded model.

Diagnosis. The sub-domain breakdown pinpoints the failure. VTabFact, whose output space is just True or False, is the only sub-domain that does not collapse, and it even rises to 62.4, while every sub-domain that has to emit a specific cell value falls to near zero (VWTQ from 27.8 to 0.6, VWTQ-Syn from 33.1 to 0.0, FinTabNetQA from 84.0 to 15.4). Together with the collapse of the span-based DocVQA and InfoVQA scores, this pattern points to a shift in the model's output distribution rather than a loss of perception. The proximate cause is a mismatch in answer style between training and evaluation: the synthetic table-QA source we trained on supplies verbose, full-sentence answers, whereas every target benchmark is scored on short cell values or spans. Fine-tuning therefore taught the model to be verbose, which exact-match and ANLS scoring penalize across the board, and the 25% replay fraction was too small to hold the line.

A corrected recipe. The diagnosis points to a concrete and falsifiable correction, which we leave to future work. First, train on short-answer table lookup whose answer length and style match the evaluation, either by filtering the source to short answers or by using WikiTableQuestions-derived short answers under the same overlap guard. Second, raise the replay fraction to 40 or 50% to protect DocVQA and InfoVQA. Third, lower the learning rate, to 5×10^{-5}, and add early stopping on a held-out development gate. Fourth, add a data-quality check before training that compares the answer-length distribution of the training mixture with that of the benchmarks; that single check

[1] Dataset: `https://huggingface.co/datasets/savoji/minivlm-tablevqa-sft`; adapter: `https://huggingface.co/savoji/qwen3.5-0.8b-tablevqa-lora`.

would have caught this failure before any GPU time was spent. The VWTQ gap identified in Section 7 still stands and remains the right target, but closing it calls for answer-style-matched supervision, not simply more table data.

9 Beyond-Accuracy Evaluation: Two Future Protocol Extensions

Our protocol already covers two axes beyond raw accuracy: diversity of correctness metrics (accuracy, ANLS, and relaxed accuracy) and the trade-off between accuracy and efficiency. We designed two further extensions, confidence calibration and robustness to input degradation and domain terminology, but we leave their execution to future work given the time and compute budget of this study. We specify both precisely below, so that each slots into the same reproducible harness without any redesign, and we present them as concrete next steps rather than speculative additions.

Calibration. We will elicit confidence in two complementary ways: verbalized confidence, where the model appends an estimate from 0 to 100% under a prompt applied uniformly to all six models, and, where the wrapper exposes them, sequence log-probabilities. Binning confidence against empirical correctness gives the Expected Calibration Error [6]. Calibration is most informative when a model is accurate but overconfident, or when it separates models that are otherwise close, and our results suggest both conditions hold for the middle of the field.

Robustness to domain terminology and degradation. We will build paired perturbations and report the accuracy drop against clean inputs. The first is image degradation, through blur, JPEG compression, and downscaling, which probes the resolution-strategy differences identified in Section 6. The second is paraphrase of the domain terminology in the questions, which tests sensitivity to surface wording. Reporting the accuracy drop on matched slices separates brittleness from baseline capability. We specify these probes here so they can be run under the same reproducible harness.

10 Limitations and Threats to Validity

Subsampling and single seed. We evaluate on $N{=}1000$ fixed-seed subsets rather than full splits, and we report point estimates without confidence intervals, so small differences, such as Qwen3.5 at 79.2 against InternVL3 at 79.4 on OCRBench, should be read as ties. **Mixed precision.** Every model runs at 16-bit, but fp16 and bf16 are not bit-identical, so precision is held constant only up to that. **Heuristic scoring.** We use the harness's string-based scorers without an LLM judge, which can under-credit verbose answers. **Latency hardware.** The built-in and custom models ran in different Colab sessions on T4 and L4 hardware; accuracy is unaffected, but cross-model latency is indicative rather than absolute, and only the ordering is robust. **Calibration and robustness** are specified but not yet measured (Section 9). **Capacity ceiling.** At 0.8B the encoder limits legibility on dense tables, which caps the gain any Part-2 method can achieve.

11 Conclusion

We can now answer the question we set out with. Between 0.45 and 0.9B parameters, VLMs are a conditionally reliable foundation for document understanding. For text-centric reading and document QA the best model, Qwen3.5-0.8B, is already credible (OCRBench 79, DocVQA 89), but for structured tabular reasoning it is not yet reliable (TableVQA 50.1). Two design lessons generalize beyond our particular models: architecture beats parameter count in this band, and the resolution and tokenization strategy is the main driver of both accuracy and efficiency. Decomposing the best model's weakness shows that its table gap is specifically one of visual lookup, not numeric reasoning, and this correction makes the improvement target precise and the adaptation testable against a criterion fixed in advance. Our first adaptation does not meet that criterion. It induces catastrophic forgetting, which we trace to a mismatch in answer style between training and evaluation, so the gap remains open. The no-regression protocol nonetheless did exactly its job: it rejected a degraded model and turned a negative result into a precise, falsifiable recipe for the next attempt rather than a misleading headline number. We release the pipeline, dataset, and scripts to support reproduction and extension.

Reproducibility and released artifacts. Code, configuration, seeds, the pinned evaluation engine, the dataset manifest, and all scripts (evaluation, sub-domain reporting, data construction, adaptation, the no-regression gate, and figure generation) are public at the repository linked above. The training mixture is released as a dataset, `https://huggingface.co/datasets/savoji/minivlm-tablevqa-sft`, and the (gate-failing) Part-2 adapter as a model, `https://huggingface.co/savoji/qwen3.5-0.8b-tablevqa-lora`, published for transparency of the negative result rather than as a recommended checkpoint. The full result tree, meaning the per-benchmark predictions, per-pair scores, logs, and summary tables for every run, is publicly mirrored on Google Drive: `https://drive.google.com/drive/folders/1xDkoQLUG3d-ymxNvvLpKryRsGe8xpLvc`.

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A Appendix

A.1 Adaptation hyperparameters

LoRA rank r=16, α=32, dropout 0; learning rate 1×10^{-4} with cosine schedule and 3% warmup; one epoch; effective batch size 16 via gradient accumulation; `max_seq_length=2048`; gradient checkpointing; vision encoder frozen; adapters on attention, MLP, and gated-delta projections; bf16 (no 4-bit). The training mixture is $\sim$70% Wikipedia-style table lookup and $\sim$25% held-out document replay, with the overlap guards of Section 8.

A.2 Qualitative examples

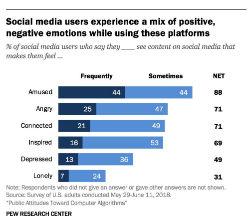
Figure 4 shows one representative example per benchmark (the diagnostic ladder), each with Qwen3.5-0.8B's correct answer. Figure 5 is a gallery of Qwen3.5-0.8B's failures on Wikipedia-style visual tables (VWTQ/VWTQ-Syn). The errors are concretely lookup mistakes, such as miscounting rows, returning a value from an adjacent column or row, or emitting a cell of the wrong type, rather than failures to read the table, consistent with the sub-domain diagnosis of Section 7.

A.3 Pipeline scripts

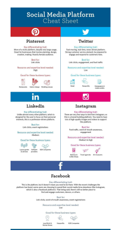
`run_eval.py` (config-driven evaluation runner; one entry point scores any registered model), `tablevqa_subdomain_report.py` (sub-domain decomposition), `efficiency_ablation.py` (precision/token-cap ablations), `build_table_sft.py` (training-mixture construction with contamination guards), `train_lora_qwen.py` (PEFT adaptation), `regression_gate.py` (pre-registered no-regression decision), `push_dataset_hf.py` (dataset release), and `make_report_figures.py` (figures).

OCRBench. *Q:* what is written in the image?
Gold: CENTRE **Qwen:** centre

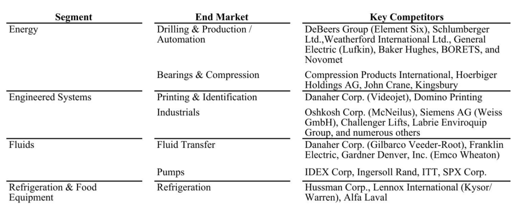
DocVQA. *Q:* What time is the 'coffee break'?
Gold: 11:14 to 11:39 a.m., 11.14 to 11.39 a . . . **Qwen:** 11:14 to 11:39 a.m.



ChartQA. *Q:* What was the 4th most popular emotion?
Gold: Inspired **Qwen:** Inspired



InfoVQA. *Q:* Which platform is suitable for software providers?
Gold: linkedin **Qwen:** LinkedIn



Segment	End Market	Key Competitors
Energy	Drilling & Production / Automation	DeBeers Group (Element Six), Schlumberger Ltd.,Weatherford International Ltd., General Electric (Lufkin), Baker Hughes, BORETS, and Novomet
	Bearings & Compression	Compression Products International, Hoerbiger Holdings AG, John Crane, Kingsbury
Engineered Systems	Printing & Identification	Danaher Corp. (Videojet), Domino Printing
	Industrials	Oshkosh Corp. (McNeilus), Siemens AG (Weiss GmbH), Challenger Lifts, Labrie Enviroquip Group, and numerous others
Fluids	Fluid Transfer	Danaher Corp. (Gilbarco Veeder-Root), Franklin Electric, Gardner Denver, Inc. (Emco Wheaton)
	Pumps	IDEX Corp, Ingersoll Rand, ITT, SPX Corp.
Refrigeration & Food Equipment	Refrigeration	Hussman Corp., Lennox International (Kysor/ Warren), Alfa Laval
	Food Equipment	Manitowoc Company, Illinois Tool, Middleby

TableVQA. *Q:* What is the end market for the Engineered Systems segment?
Gold: Printing & Identification, Industrial . . . **Qwen:** Printing & Identification Industrials

Figure 4: Representative examples across the benchmark suite (the diagnostic ladder), each with Qwen3.5-0.8B's (correct) answer. Colours: gold answer (green), model answer (green when correct).

Round	Pick	Name	Position	College
1	9	Brian Urlacher	S	New Mexico
2	39	Mike Brown	S	Nebraska
3	69	Dez White	WR	Georgia Tech
3	87	Dustin Lyman	TE	Wake Forest
4	125	Reggie Austin	DB	Wake Forest
6	170	Frank Murphy	WR	Kansas State
6	174	Paul Edinger	K	Michigan State
7	223	James Cotton	DE	Ohio State
7	254	Michael Green	S	Northwestern State

VWTQ *Q:* how many players attended wake forest university before being drafted?
Gold: 2 **Qwen:** 1

Year	Wins (majors)	Earnings ($)	Rank
1985	0	0	-
1986	0	113,245	77
1987	1	297,378	33
1988	0	156,068	75
1989	0	278,760	44
1990	1	537,172	20
1991	1	686,361	8
1992	3	1,191,630	2
1993	2	777,059	12
1994	0	474,219	33
1995	1	1,111,999	6
1996	1	1,211,139	7
1997	2 (1)	1,635,953	3
1998	1	1,541,152	11
1999	0	2,475,328	3
2000	0	2,337,765	9
2001	1	3,169,463	5
2002	0	2,056,160	21
2003	4	6,081,896	3
2004	0	3,075,092	10
2005	0	2,658,779	13
2006	1	2,747,206	16
2007	0	1,016,489	96
2008	1	1,695,237	48
2009	0	1,622,401	52
2010	0	1,214,472	73
2011	0	1,056,300	88
2012	0	989,753	100
2013	0	303,470	165
Career*	**20 (1)**	**42,511,946**	6

VWTQ *Q:* how many years did davis love iii rank in the top 10, but not have a pga tour win?
Gold: 3 **Qwen:** 1991

Year	Competition	Venue	Position	Event	Notes
2001	World Youth Championships	Debrecen, Hungary	4th	400 m	
2002	Asian Games	Busan, South Korea	2nd	4x400 m relay	
2005	Asian Championships	Incheon, South Korea	2nd	4x400 m relay	
	Universiade	Izmir, Turkey	6th	4x400 m relay	
2006	Asian Games	Doha, Qatar	2nd	4x400 m relay	
			1st	400 m	
	World Cup	Athens, Greece	7th	400 m	
2007	Universiade	Bangkok, Thailand	1st	400 m	
2011	Universiade	Shenzhen, China	–	400 m	DQ

VWTQ *Q:* what is the total number of races olga tereshkova has won.
Gold: 2 **Qwen:** 1

Position	Player	Transferred From	Date
GK	Fajar Sarib	Geylang United	1 January 2010
GK	Hafez Mawasi	Balestier Khalsa	1 January 2010
DF	Winston Yap	Sengkang Punggol	1 January 2010
DF	Kazuki Yoshino	Albirex Niigata (S)	1 January 2010
DF	Sahairi Ramri	Balestier Khalsa	1 January 2010
MF	Syed Karim	SAFFC	1 January 2010
MF	Mohd Noor Ali	Geylang United	1 January 2010
MF	Guntur Djafril	SAFFC	1 January 2010
MF	Sazali Salleh	Sengkang Punggol	1 January 2010
FW	Laakkad Abdelhadi	Free Transfer	1 January 2010
FW	Rizawan Abdullah	Balestier Khalsa	1 January 2010
MF	Rachid Lajane	Raja Al Hoceima	26 February 2010

VWTQ-SYN *Q:* who is the next added player after january of 2010 to the roster?
Gold: Rachid Lajane **Qwen:** Guntur Djafril

Year	State/Territory Men's Di vision	State/Territory Wome n's Division	Major Centres Division	Community Division	Women's Division
2012	Queensland	New South Wales			
2012	New South Wales	New South Wales	Darwin	Brothers in Arms	Bush Potatoes
2011	New South Wales	New South Wales	Maranoa Murris	Gap Angels	Bush Potatoes
2010	Western Australia				
2009	Queensland		Alkupitja	Tangentyere	New South Wales
2008	Queensland		Katherine	Cooktown	New South Wales
2007	New South Wales		Alkupitja	Cat Tigers	CGA Cougars
2006	Queensland		Alice Springs	Melville Island	Darwin
2005	Queensland		Alice Springs	Alkupitja	Darwin
2004	Queensland		Alice Springs	Normanton	Tennant Creek
2003	New South Wales		Darwin		
2002	Northern Territory		Darwin		
2001	Tasmania				

VWTQ-SYN *Q:* what column comes before the major centres division?
Gold: State/Territory Women's Division **Qwen:** State/Territory Men's Division

Rank	Name	Nationality	Time	Notes	Points
	Adam Kszczot	Poland	1:46.50		**12**
	Jeff Lastennet	France	1:46.70		**11**
	Gareth Warburton	Great Britain	1:46.95	SB	**10**
	Mario Scapini	Italy	1:47.20	PB	**9**
	Anis Ananenka	Belarus	1:47.29		**8**
	Oleh Kayafa	Ukraine	1:47.42		**7**
	Joni Jaako	Sweden	1:47.61	SB	**6**
	Robin Schembera	Germany	1:47.79		**5**
	Ivan Tukhtachev	Russia	1:48.27	SB	**4**
10	Antonio Manuel Reina	Spain	1:48.56		**3**
11	António Rodrigues	Portugal	1:50.45		**2**
12	Milan Kocourek	Czech Republic	1:59.28		**1**

VWTQ *Q:* who is ranked directly before jeff lastennet?
Gold: Adam Kszczot **Qwen:** Anis Ananenko

Figure 5: Failure gallery on Wikipedia-style visual tables (VWTQ/VWTQ-Syn), the measured weak sub-skill. The errors are *lookup* mistakes— miscounting rows, returning an adjacent column/row, or emitting a cell value of the wrong type—not failures to read the table. Gold (green) vs Qwen (red).